

① sources

② sloganize

③ embed

④ retrieve

⑤ judge

INFORMAL

LaTeX stmt

arXiv / blueprint
informal graph

LLM

slogan

one-line plain-English
description of the statement

embed

slogan vector
 $\mathbf{v}_I \in \mathbb{R}^d$

shared index

single embedding space
cosine similarity

one index, both lanes

candidate match

formal $\rightarrow$ informal
rank-1 sweep
rank-1 neighbor
sim. score $s = \cos$

label

LLM-as-judge • GPT-5.4

exact

same result, no caveat

inexact

same result up to a generalization,
special case, or one direction of an
iff

wrong

different results despite high
embedding similarity

exact / inexact = match *wrong = non-match*

Lean decl

Mathlib / projects
formal graph

LLM

slogan

one-line plain-English
description of the statement

embed

slogan vector
 $\mathbf{v}_F \in \mathbb{R}^d$